Data Engineers vs Data Analyst vs Data Scientists

Prepared for

Dr. Aryati Bakri

Senior Lecturer, Faculty of Computing, University Technology Malaysia

By

Mohamed Alif Fathi bin Abdul Latif[1] and Iman Abadi bin Mohd Nizwan[2]

Students, Faculty of Computing, University Technology Malaysia

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I. Executive Summary

This paper explores how data professional roles such as data engineers, data scientist and data function in an organization. The paper aims to identify how each role differs based on their roles, responsibilities, technologies used and skills required. A review was done on how these roles became important and essential in an organization as more of them are moving towards big data for data-driven decisions. There are also findings on the key skills needed for data professionals in order to execute their task effectively and how emerging technologies help organizations take advantage of big data more effectively. The paper concludes on the importance of these roles with some recommendations to enable more advancement in big data and data analytics.

II. Introduction

In this era of modernization, organizations produce massive amounts of data from their systems. With this large amount of data, the organization makes data-driven decision-making to grow and stay ahead from competitors. To effectively utilize the data, companies make data pipelines to enable data-driven decisions.

Data pipelines are chains of operations that manipulate data, allowing data to flow smoothly and automatically from sources to destination for insights and analytics (Munappy et al., 2020). The whole data pipelines systems require roles to effectively make the data-driven decision-making. The roles included are data engineer, data analyst and data scientist. Each of these roles have different jobs in which a data engineer created and designed the data pipeline. Once the data is processed and ready, data analysts and data scientists use the information to analyse and find insights for the company to grow. However, these roles are often misunderstood like data analyst and data scientist have the same job scope. This confusion can lead to inefficiencies and unclear responsibilities leading to job redundancy.

Because of these problems, understanding the difference between these roles is crucial for organizations aiming to optimize their data pipeline systems. Clear role definitions can improve collaboration between teams and ensure accurate expertise in data pipelines workflow for accurate decision-making.

Therefore, this report aims to compare the roles of data engineer, data analyst and data scientist. The report will define each role, determine their responsibilities, tools and technology used, skill sets and show how they interact within a data pipeline workflow. Additionally, it will discuss challenges, future trends and provide recommendations for organizations to better structure their data teams.

III. Background/Literature Review

Recent studies highlight the importance of the data-driven approach in their systems. As more companies rely on data pipelines systems, data engineer, data analyst and data scientist roles become more demanding. According to Thammasudjarit (2024), the roles of data engineers, scientists, and analysts are important to improve patient outcomes and operational efficiency in the healthcare industry. Data engineers first create infrastructure by building robust and secure data pipelines with multiple data sources integration. Data scientists then focus on methodologies of modeling techniques to develop proof-of-concept solutions for complex medical problems while data analysts interpret this processed information to provide actionable insights that optimize patient care and hospital management (Thammasudjarit, 2024).

Another study shows emerging cloud technology reshaped data engineering. According to Rosa et al. (2025), cloud platforms offer scalability, flexibility and durability to the system making it important for data pipelines systems. Many organizations are transforming their data storage and processing which is a part of data pipelines to cloud infrastructure by using services from providers such as AWS (Amazon Web Service), Microsoft Azure and Google Cloud Platform. This evolution makes data engineers expected to have strong skills for cloud-based technologies including data warehousing like Snowflake and data processing like Apache Spark. For data scientists, cloud platforms like AWS Glue and AWS EMR are really helpful for machine learning development by converting the heterogeneous data from various sources in different formats to a single homogenous format (Gupta & Sharma, 2023).

In addition, the existing paper also highlights that even the data specialists have different tasks, skills needed for them often overlapping. According to Han and Ren (2024), data specialists not only need technical skills, but also soft skills like work in collaboration and communication skills. For instance, data engineers not only build data pipelines but also need to understand business goals for better understanding of data flow in whole pipelines to make sure it is actually useful. Furthermore, Kobets et al. (2024) found that some roles are more focusing on analysis skills like SQL while others are focusing on tools like Docker to make and maintain a platform for data-driven systems. This means that a data scientist in one company might do some data engineering work, while a data engineer might spend time on analysis which depends on the tools the company uses.

IV. Role Overview

The roles of data engineer, data analyst, and data scientist have different purposes for the whole data pipeline system. It is essential to understand each role in an organization to effectively create and manage data pipelines and make data-driven decisions.

A. Data Engineer

Data engineering is the most fundamental role that must be established before other roles can be effectively utilized. A data engineer is responsible for creating pipelines that ingest and integrate data from multiple sources, transforming them to produce high-quality data suitable for data analytics and machine learning (Thammasudjarit, 2024). They are also responsible for ensuring that the pipeline is scalable to accommodate and process large amounts of data using technologies such as cloud computing for better flexibility and cost-efficiency. Cloud computing offers various services for each component of a data pipeline. Many cloud providers offer extensive services to produce an end-to-end data pipeline from cloud storage solutions, data warehousing, and data processing tools.

B. Data Analyst

A data analyst uncovers insights and trends from historical data so that stakeholders can determine the next course of action for business growth. A data analyst must be skilled in data visualization, reporting, effective storytelling, and understanding business goals (Thammasudjarit, 2024). Data visualization skills are essential, as they represent data in graphs and charts that can be easily interpreted and understood. Reporting on the other hand focuses on structured writing with information updates on key metrics that drive the business goals. Effective storytelling allows data analysts to facilitate the flow of their presentation to their stakeholders so that it is engaging and easily understood with the support of data visualization skills and reporting skills. This skill allows the stakeholders to have a clear grasp of the situation to determine the best action to take.

C. Data Scientist

While data analysts focus on the past, a data scientist focuses on the future by making predictive models that predict future trends. A data scientist's skill set includes ML modeling, experimentation, statistics and research (Thammasudjarit, 2024). According to Afsharian (2024), a typical data science team workflow has six total steps, which are data understanding, data acquisition, data preparation, data modeling, model validation and communication of results. Data understanding is the key part of the workflow where the data scientist need to have domain knowledge and expertise in order to define the projects objectives. Data acquisition and preparation is standard in any data roles, determining data to be used and preparing the data to train the model. Data modeling is considered the core part of a data scientist's work. In this stage, data scientist develops an ML model that predicts future trends from complex datasets. The model is then evaluated using statistical knowledge to determine the accuracy of the model produce before it can be deployed. Once the model is deployed, the model's performance will receive feedback to make improvements.

V. Tools and Technologies

There are many available tools for data engineers, data analysts and data scientists to use for their jobs. These tools are different based on the responsibility of each role. Some tools may overlap but each role mainly focused on specific technology that aligns with their core functionality in data pipelines.

Data engineers work with tools that are designed to build data pipelines infrastructure that support data ingestion, data processing, storage and automation. Data analyst more to tools that support querying, reporting and visualization. Data scientists use tools that support programming to develop advanced analytical models and predictions using machine learning.

Table 5.1 Comparison of Tools and Technologies for each roles

Role	Category	Tools
Data Engineer	Programming	Python, Java
	Data Ingestion	Apache Kafka, AWS Glue
	Data Processing	Apache Spark, Hadoop
	Data Storage	Amazon S3, Azure Blob Storage, Azure Data Lake Storage
	Data Warehousing	Amazon Redshift, Snowflake, Google BigQuery
Data Analyst	Query Language	SQL
	Data Analysis	Microsoft Excel, Excel Solver
	Data Preprocessing	Python (Pandas)
	Data Visualization	Power BI, Tableau
Data Scientist	Programming	Python, R
	Machine Learning	Scikit-Learn, TensorFlow, Pytorch
	Development Tools	Jupyter Notebook
	Big Data Tools	Apache Spark (MLlib)

VI. Data Pipeline Workflow

A data pipeline is a foundation of collaboration for data engineer, data analyst and data scientist. According to Beevi (2025), a scalable workflow follows four stages which are data ingestion, data processing, data storage and data visualization to ensure data flow accurately and consistently. A figure below shows how these stages interconnected.

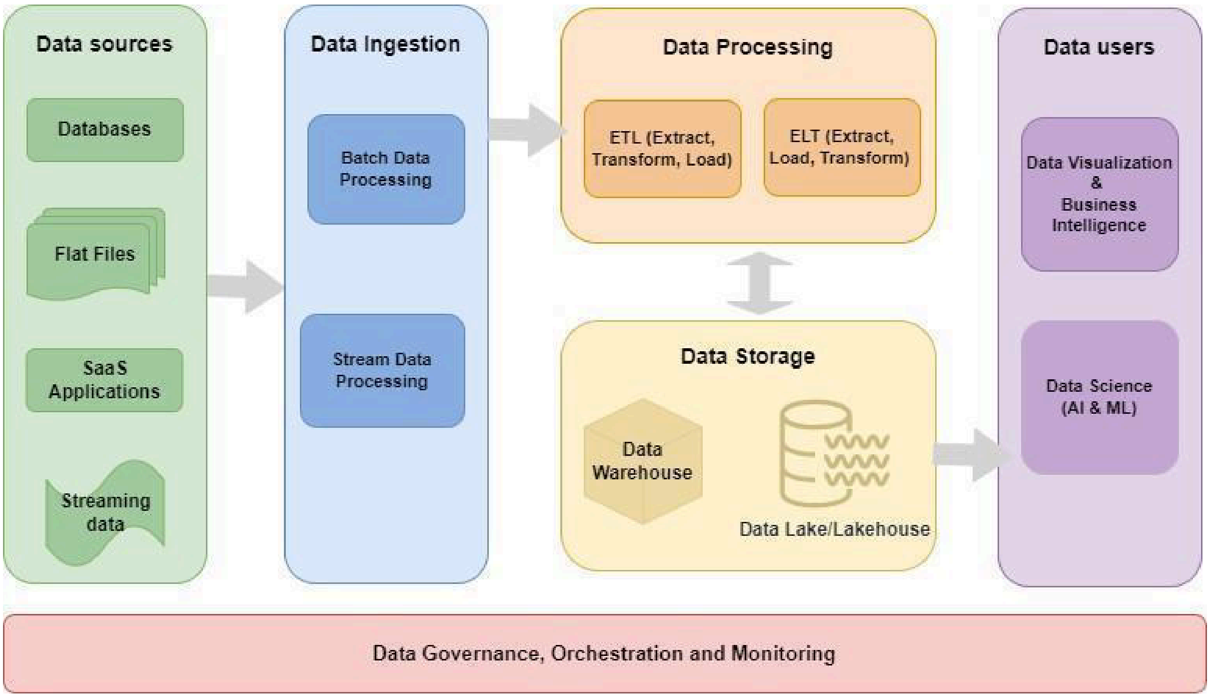

Figure 6.1 Data Pipeline Components

It begins with the first stage which is data ingestion. This stage is the start point to where data from industry and other different data sources are collected to be used for the whole process of pipeline. The example of data sources are sensors, social media or the system databases. Tools like Apache Kafka can be used to stream data at high speed.

The next stage is data processing. Once the data is collected, the data need to be cleaned and organized. There are two types of data processing which are batch processing and stream processing. Stream processing is for continuous processing with low latency while batch processing handles the process with scheduled intervals where immediate result is not prioritized.

After processing, the data need to be stored in a way that is easy to access. Data lake is used to store unstructured and semi structured data while data warehouse is used to store structured data. Data lake is needed for data scientists because of raw data that can be used for machine learning while data warehouse is needed for data analysts for making analysis based on structured data.

The final stage is data visualization. This stage is crucial for data analysts because the information gained in this stage is used to find business insight for companies by making dashboards and reports. For data scientists, this stage provides high-quality data to run machine learning models and develop analytical solutions.

The data pipelines system must also be designed for fault tolerance and scalability. This means that if one part of the pipeline has problems, the system can recover without losing data and the pipeline can handle more data in the future as the company continues to grow (Beevi, 2025).

VII. Challenges and Future Trends

One of the main challenges for professional data roles in big data is that most companies struggle to process large amounts of data (Goyal et al., 2020). Companies lack funds to invest on other data sources, big data resources and also individuals with related domain expertise. The lack of funds also reflect on companies' limited abilities to invest on softwares and resources that can scale with the amounts of data available and deploy the data solutions. Furthermore, there is also a shortage of analytical talent which is defined by individuals who can turn the raw information to some meaningful information (Goyal et al., 2020). Another challenge is on data management to ensure security and privacy. The presence of data is the very reason for the existence of the data analytics fields. However, there are still some concerns on data security as there are no fully secured data infrastructures especially on sensitive data. Even if there are efforts on improving the security, there are also external parties that will try to overcome and break through the security measures.

Future trends for these data roles are the implementation of AI. AI automation has given an impact on data roles such as data engineer and data scientist. A paper by Shivpuja (2025) described the use of AI-powered tools that can generate pipelines automatically, optimize database schemas and suggest improvements in pipeline architecture. Data scientist on the other hand has changed its main focus from technical execution to a strategic analysis oriented role. They focus more on the impacts on business rather than model development. This allows these roles to be strategic planning and optimization rather than technical practices.

The presence of AI has also contributed to the emergence of new and hybrid critical roles like AI Solutions Architect, AI Behavior Analyst, AI Bias Auditor, AI Workflow Engineer, and Data Ecosystem Manager (George et al., 2025). An AI Solutions Architect for example, plans an organization's AI implementations. They provide how business solutions can be improved with the help of AI. This role requires technical skills such as knowledge in cloud computing architectures, data processing systems and AI frameworks which can be found in skills required for both data engineers and data scientists. Another example is an AI Behaviour Analyst whose main role is to improve AI behaviours by analyzing training datasets or model architecture to improve its accuracy which is similar to that of a data scientist.

VIII. Recommendations

In order to effectively take advantage of big data and data analytics, organizations should take actions on utilizing professional data roles data engineers, data analysts, and data scientists. Data roles should be clearly defined in terms of roles and responsibilities to avoid confusions and overlapping tasks. Many organizations expect data professionals to perform multiple data related tasks that are out of their field of expertise, especially growing organizations. For example a data analyst doing data engineering related tasks. This can lead to inefficiencies and reduction in productivity.

Furthermore, invest in modern data infrastructure and emerging technologies. They should invest in a cloud-based platform that promotes scalability and flexibility to manage high volumes of data and support advanced analytics. Additionally, implement AI to assist data roles in automating their tasks. Automation allows them to automate repetitive tasks to focus on strategic planning and system designs rather than worrying about the technical implementation.

Finally, encourage continuous skill development among data professionals as the field of data analytics and data engineering is continuously evolving as new technologies and tools are developed. Organizations need to invest in upskilling them through training programs, professional certificates and workshops. This will keep their employees up to date with emerging technologies such as machine learning and cloud-platforms. This will directly impact the organizations as their employees will apply the knowledge they had learned in building a more robust data architecture and analytics capabilities that drive their business goals.

IX. Conclusion

In conclusion, data professionals are essential for organizations to make data-driven decisions. In the age of information, data is valuable as it provides business insights from trends and patterns in order to achieve business goals. Data analysts, data engineers and data scientists can help in achieving business goals by implementing their respective skillsets to create an end-to-end data workflow from transforming raw data into actionable insights. Although there are challenges, such as lack of funds, data talents and data security concerns, there are still solutions that can be done such as upskilling and investing in emerging technologies such as AI and cloud-platforms. These can effectively improve the workflow of data professionals to improve efficiency and increase productivity.

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